BUSINESS PRESENTATION

ChurnGuard

Predict churn. Prioritize customers.

Test retention offers before investing at scale.

Customer Churn Prediction using Machine Learning

Evidence: held-out model evaluation and notebook-based ROI scenarios

Audience: CMO | COO | Data Science Manager

Executive summary

What the analysis establishes - and what still requires a pilot

16.8% churn

About 17 in every 100 customers churned in the analyzed data.

Strong prediction

XGBoost achieved 94% recall and 86% precision on 2,252 test records.

Focused intervention

372 Critical/High accounts hold 78.60% of estimated monthly revenue exposure.

EXECUTIVE DECISION

Launch a controlled pilot using low-cost outreach first; treat all retention savings and ROI as scenario estimates.

Business problem and customer patterns

Why a targeted retention approach is more efficient than blanket incentives

Customer attrition

Observed churn: 16.8%. At that rate, about 336 of 2,000 accounts would churn.

Customer characteristics

Super and Regular Plus represent roughly 70% of customers; payment method also varies.

Predictive drivers

Top importance: account segment 20.55%, payment 14.59%, tenure 11.85%.

INTERPRETATION

Importance shows model influence, not proof that changing a feature will prevent churn.

End-to-end solution

From historical customer data to action in the Prediction App

1. Understand

Define churn

2. Explore

Inspect patterns

3. Prepare

Clean and encode

4. Train

Compare models

5. Evaluate

Held-out testing

6. Prioritize

Risk segments

7. Plan offers

Cost + scenarios

8. Prediction App

Score + export

Historical data -> Predictive model -> Customer risk list -> Controlled retention campaign

Measured model performance

Actual XGBoost results on 2,252 held-out customer records

Recall 94%

Finds about 94 of every 100 actual churners; 6 may be missed.

Precision 86%

About 86 of every 100 flagged customers actually churned.

Accuracy 96%

Overall, about 96 of 100 churn/non-churn classifications were correct.

F1-score 0.90

Balances precision and recall into one score; useful when both missed churners and unnecessary outreach matter.

Why multiple scores?

Accuracy alone can hide poor churn detection when non-churners are the majority; use recall, precision and PR-AUC together.

Other measured scores - explained

Ranking quality, precision-recall trade-off, and repeatability

PR-AUC 0.96

Summarizes precision versus recall across decision thresholds. Particularly relevant when churners are less common.

ROC-AUC 0.99

Measures how well the model ranks churners above non-churners across thresholds. Near 1.0 indicates excellent separation.

Cross-validation PR-AUC 0.9481

Performance averaged across validation folds during model selection; supports consistency beyond one split.

F1-score 0.90

The harmonic mean of precision (0.86) and recall (0.94). It penalizes a model that performs well on only one.

Where churn exposure is concentrated

Held-out sample: 2,252 accounts; probability-weighted monthly revenue

Risk group	Accounts	Account share	Monthly exposure	Exposure share
Critical	**315**	**13.99%**	**2,196.30**	**68.36%**
High	57	2.53%	329.08	10.24%
Moderate	118	5.24%	273.54	8.51%
Low	1,762	78.24%	414.00	12.89%
Critical + High	372	16.52%	2,525.38	78.60%

19 of the 20 highest-ranked test accounts actually churned. Revenue uses dataset units, not a confirmed currency.

Prediction App: from scoring to decisions

Operational use case - single customer assessment or batch account prioritization

Single prediction

Enter customer attributes, receive churn probability, risk segment and suggested next action.

Batch prediction

Upload CSV/XLSX accounts; score and sort by risk; export prioritized lists to CSV or Excel.

Risk-led actions

Critical: immediate outreach; High: proactive offers; Moderate: preventive service; Low: routine engagement.

GOVERNANCE

Review data quality, privacy, model drift and campaign eligibility before using scores for customer treatment.

Retention success assumptions

Scenario success rates are assumptions - not observed customer responses

Scenario	Critical success	High success	Interpretation
Conservative	**15%**	**20%**	**Lower response / downside**
Base Case	30%	40%	Planning reference
Optimistic	45%	60%	Upside sensitivity

SHARED FINANCIAL ASSUMPTIONS

12 months of retained revenue, 30% contribution margin; offer costs calculated from segment monthly revenue.

Offer costs and base-vs-optimistic ROI

Illustrative ROI by segment and offer - not realized returns

Offer	Cost rate	Critical Base	Critical Opt.	High Base	High Opt.
Personalized outreach	**5%**	**1,935%**	**2,953%**	**1,891%**	**2,887%**
Loyalty reward	10%	918%	1,426%	896%	1,393%
Discount voucher	15%	578%	918%	564%	896%
Cashback incentive	25%	307%	511%	298%	498%
Premium benefit	60%	70%	154%	66%	149%

ROI = (estimated retained contribution - offer cost) / offer cost. Rounded values; optimistic values are not the base case.

What do Base and Optimistic mean in value?

Example: 5% personalized outreach across the Critical and High groups

Group	Campaign cost	Base net benefit	Optimistic net benefit
Critical (315)	**116.55**	**2,255.45**	**3,441.46**
High (57)	23.80	450.08	687.01
Combined (372)	140.35	2,705.53	4,128.47

HOW THE VALUE IS ESTIMATED

Expected retained contribution = monthly revenue exposure x success rate x 12 x 30% margin.

Net benefit = retained contribution - campaign cost. Amounts are dataset revenue units.

Conclusion and recommended next steps

A promising prioritization tool - validate financial impact before scale

1. Deploy carefully

Use the Prediction App to produce reviewed Critical/High lists and assign outreach owners.

2. Run a pilot

Randomize eligible accounts into outreach and control groups; compare low-cost outreach with loyalty.

3. Measure outcomes

Track actual retained customers, incremental revenue, campaign spend, realized ROI and fairness.

FINAL RECOMMENDATION

Prioritize the 372 highest-risk test accounts as a proof of concept, but do not claim prevented churn or ROI until measured in a live controlled pilot.